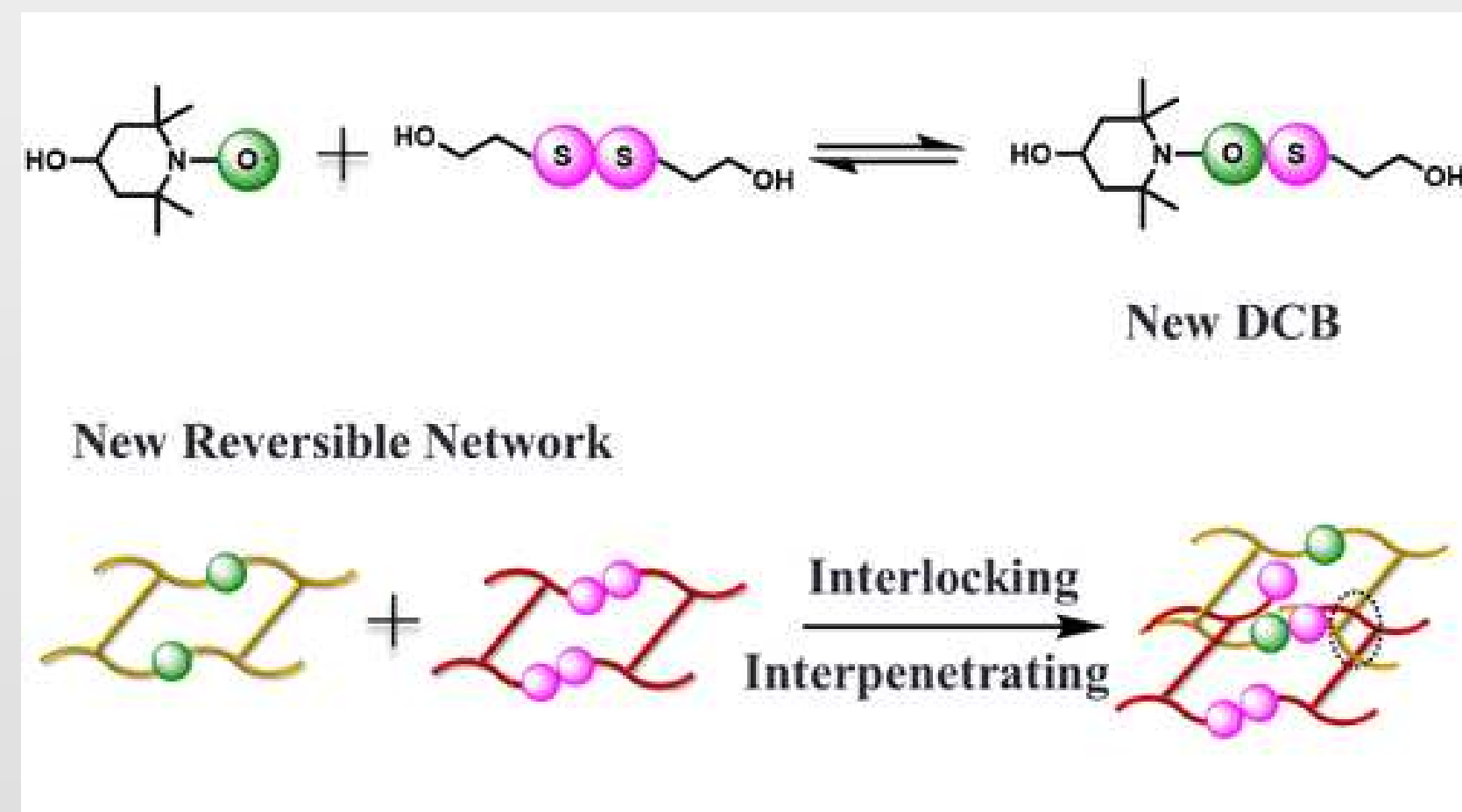


Determining the Effect of Dynamic Covalent Bonds on the Tensile Strength of Hardened Epoxy Resin

CHEM

Introduction

Epoxy is a strong adhesive used in **many high stress situations**. It works by **combining two different components** to create a **polymer**, or a material made of repeating smaller chemical structures. **Dynamic covalent bonds** are bonds in an equilibrium state of **breaking** and **reforming** (Roh et al., 2024), which gives them the ability to **modify certain characteristics** of polymers, such as strength, self-healing factors, or elasticity (Ngyuen & Lee, 2024).



Example demonstrating how dynamic bonds function
Image taken from Li et al. (2025)

Due to the **intense situations** epoxy is in, sometimes it can **fail**, which can cause **critical damage** to machines or other structures. Specifically, the pulling force it receives is substantial, meaning the epoxy requires a **higher tensile strength**.

Purpose

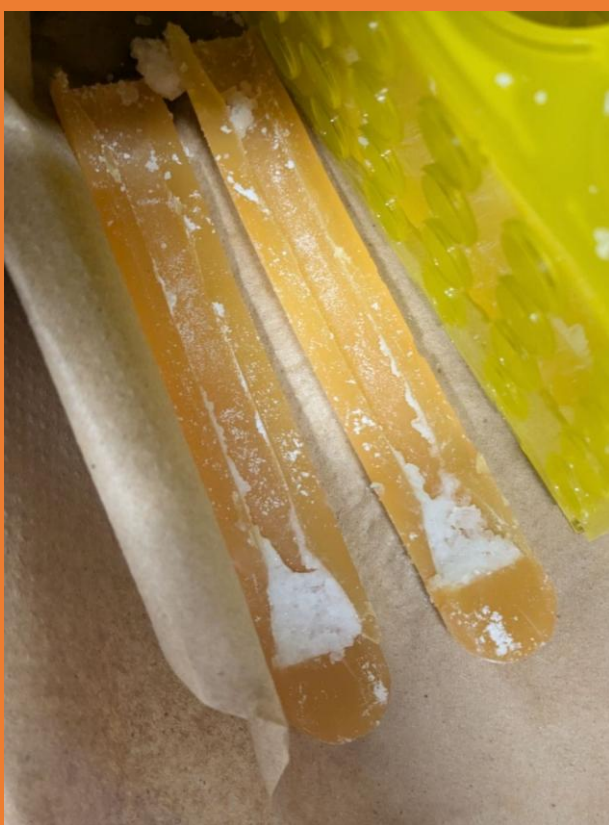
One possible solution for the problem of epoxy failing is **dynamic covalent bonds**. By synthesizing these bonds with the epoxy, it may be possible to **increase its tensile strength**, which has yet to be tested with a mechanical grade epoxy like the one I plan to use. I **hypothesized** that by **adding the dynamic bonds**, the **tensile strength** would **increase** based on dynamic bond interactions with **other polymers**.

Methodology

Variables

IV: Presence of dynamic bonds
DV: Tensile strength of epoxy

1. Create **mold shape** for epoxy
 - 3d print ideal shape and use silicone to form a mold around it



Mold for the Epoxy

2. Create **control group** of epoxy
 - Heat Bisphenol A Diglycidyl Ether (base component of epoxy) (BADGE) until it melts
 - Mix BADGE with Polypropylene Glycol diamine (curing agent) at high temperature
 - Keep hot in an oven for 4 hours

3. Create **dynamic bond curing agent**
 - Stir dimethyl 3,3-dithiodipropionate (contains dynamic bonds) with Polypropylene in hot oil bath for 6 hours
 - Use FTIR to verify

4. Create **dynamic bond incorporated epoxy resin**
 - Heat BADGE until it melts
 - Mix BADGE with dynamic bond curing agent at high temperature (Earls et al., 2007)
 - Use FTIR to verify
 - Heat in an oven for 4 hours



Demonstration of Step 4

5. Test **tensile strength**

- Use MARK-10 Force Stand to test tensile strength of control and dynamic infused epoxy resin



MARK-10 Force Stand
<https://mark-10.com/>



FTIR from one trial

Results

I am **currently still collecting data**, as I have been given the **wrong chemical** from the inventory while trying to collect data, so I have no results to share currently. However, I have **created my mold**, done **FTIR** on my **past trials**, and was able to successfully make my **control group** using my new curing method. Unfortunately, there was not enough time to collect results, but I **expect** the **tensile strength to increase** with the dynamic bonds due to other types of dynamic **bonds showing similar results** in the past.



Epoxy made with new synthesis method

Conclusion

While mislabeling of chemicals and time constraints stopped me from being able to test the dynamic bonds, I was able to show that my **method of curing BADGE** to include the dynamic bonds is **viable**, as I created a **piece of epoxy resin** with an untested curing agent for BADGE. **In the future**, this process could be tested with the **dynamic bonds**, as the process has now been **finalized**.

References

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